

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 | SUB-STRAND 1.5: MOMENTS OF EQUILIBRIUM | FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Welcome to your Final Explanation! Throughout this unit you have investigated the Toppling Jiko Stool: a three-legged stool stood firm under a sufuria of ugali, but toppled when a full 20-litre jerrycan (weight 200 N) was placed at one corner and nudged, while a wider, lower four-legged stool with the same jerrycan at its corner only rocked and returned upright. Use your evidence from Lessons 1 to 6, your beam-balance and metre-rule data, your centre of gravity investigations, your cumulative model and your Driving Question Board to answer each prompt below. Write in full sentences, use correct physics words, show every step of any calculation with units, and refer to evidence you actually observed. Draw neat labelled diagrams where asked. Take your time, and think about how each answer helps answer our Driving Question: Why do some objects stay balanced while others topple, and how can we use the physics of turning forces to predict and control whether an object will be stable or fall?

Part 1: Turning Effects and Moments

In Lessons 1 and 2 you saw that the jerrycan caused toppling at the corner of the Jiko stool but did not cause it when the load was placed elsewhere. (a) What is a moment of a force, and what is the formula and SI unit used to calculate it? (b) Using your metre-rule and pencil-pivot activity, describe what you observed when you moved the 500 g mass further from the pivot, and explain why this happened even though the mass did not change. (c) Suppose the 200 N jerrycan acts at a perpendicular distance of 0.30 m from a pivot leg. Calculate its moment about that leg. Then calculate the moment if the jerrycan were moved closer to the pivot, to a perpendicular distance of 0.10 m. What does this comparison tell you about why the position of the jerrycan matters?

Write your answer here:

Part 2: Centre of Gravity and the Tipping Point

In Lesson 3 you used a plumb line and cardboard cut-outs, and tested blocks to see when they topple. (a) Define centre of gravity and describe how you found it for an irregular cardboard shape. (b) Using what you saw with the tall narrow block and the wide low block, state the condition that decides whether an object topples. (c) Explain, using centre of gravity, why the three-legged Jiko stool toppled when the jerrycan was placed at its corner. (d) Name the three states of equilibrium. Then decide which state the wider four-legged stool was in when it was nudged with the jerrycan at its corner, and give a reason based on what you observed.

Write your answer here:

Part 3: The Principle of Moments

In Lesson 4 you balanced a metre rule on a pencil pivot (like a bao plank on a log) using bags of beans and coins and recorded clockwise and anti-clockwise moments. (a) State the Principle of Moments. (b) A bao plank is balanced on its pivot. On the left, a child's weight of 300 N sits 1.0 m from the pivot. On the right, a second child sits 1.5 m from the pivot. Calculate the weight of the second child needed for balance, showing your working. (c) A stool begins to tip about the edge of its base nearest the jerrycan. Explain, using moments about that tipping edge, what decides whether the stool topples or returns upright, and use this to explain why the wide four-legged stool did not topple with the same jerrycan at its corner but the narrow three-legged stool did.

Write your answer here:

Part 4: Stability in Design

In Lesson 5 you tested tall cardboard boxes with clay and widened the base with flaps, and analysed a crane incident in Nairobi. (a) Describe what you observed about the critical tipping angle when clay was at the top of the box compared with the base, and when the base was widened. (b) State two ways to make an object more stable and explain each using the idea of centre of gravity and base of support. (c) Explain why the crane in the case study toppled. (d) Name one real structure or career where engineers or workers use these ideas and describe how.

Write your answer here:

Part 5: Explaining and Predicting with the Jiko Stool

Now bring everything together to answer the Driving Question. (a) Write a complete explanation of why the three-legged Jiko stool toppled with the 200 N jerrycan at its corner while the wider four-legged stool only rocked and returned upright. Link the ideas you used across the unit into one cause-and-effect chain, and explain both stools. (b) Predict what would happen if the jerrycan were placed at the centre of the three-legged stool, and justify with physics. (c) Suggest two changes a Kenyan market trader could make to a stool or a stack of goods, such as sacks of maize or crates, to stop it toppling, and explain why each works. (d) How does this unit show that Physics is relevant in daily life and that forces determine the stability of everyday objects?

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Moment calculations and the Principle of Moments	Correctly states $M = F \times d$ and the Principle of Moments, calculates all values with correct working and units (N·m, N), and interprets what the results mean for the stool.	States the formulas and gets most calculations right, with minor errors in units or working, and gives limited interpretation.	Gives incomplete or incorrect formulas, calculation errors, or missing units, with little or no interpretation.
Use of centre of gravity and base of support	Accurately defines centre of gravity and states the toppling condition (vertical line outside the base of support), and applies it precisely to both stools and the crane.	Defines the terms and states the condition mostly correctly, applying it to the stool with some gaps.	Confuses centre of gravity with other ideas, or does not link the vertical line to the base of support.
Evidence from the investigations	Refers specifically to observations from the lessons (500 g mass on the metre rule, plumb line, clay at top or base, flaps, 200 N jerrycan, crane case) to support every claim.	Refers to some observations from the lessons but some claims are not backed by evidence.	Makes general statements with little or no reference to what was observed.
Causal explanation of the anchoring phenomenon	Gives a complete cause-and-effect chain linking moments, shift of the combined centre of gravity, the base of support and equilibrium to explain both stools, and makes a correct prediction.	Explains the phenomenon using most key ideas but the chain of reasoning has a missing link.	Describes what happened without explaining why, or uses ideas incorrectly.

Application and communication	Gives realistic Kenyan applications with correct reasoning, uses scientific vocabulary correctly throughout, and presents clear, well-organised answers.	Gives reasonable applications and mostly correct vocabulary, with some unclear parts.	Applications are missing or unrealistic, vocabulary is often incorrect, and answers are hard to follow.
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